

Reflection: Portfolio Artifact 4

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For my fourth portfolio artifact I decided to look into model bias and fairness in a classification system trained on real data. This is a fantastic piece for my portfolio since it exhibits a skill that my previous artefacts did not have. While my first three publications showed I could develop predictive models, clean and engineer data, and articulate technical results, they lacked the ethical and evaluative aspects of AI work that firms increasingly ask practitioners to understand. This artifact shows how I think about both the validity of a model and how well it performs over various subgroups. I updated this artifact for my professional portfolio to make it seem different from the one I used in class. I updated the start to eliminate any mentions of the course and instead point to a real-world example: the need for fairness audits when models affect choices that are personal (loans, employment, or access to healthcare, for example). This framing will allow viewers unfamiliar with the academic background of the assignment to quickly grasp the importance and stakes of the effort (Alelyani, 2021). I also rearranged the presentation so that it was easier to read. Portfolio explanation A quick explanation at the beginning of the portfolio explains the problem, how I approached it, and my key discovery, so a hiring manager or recruiter skimming the portfolio may understand its worth in less than a minute.

I have included a more detailed technical breakdown below for a technical reviewer or interviewer to fully analyze my approach. That includes the measurements I employed to assure fairness, and how I assessed differences across groups. I made the words less scholarly and more businesslike, enhanced the look of my images and added labels that were easier to understand. In the process of choosing and refining this artifact, I was reminded of a point that is made throughout the portfolio: high quality academic work is not necessarily good quality professional work. When creating high-quality portfolio items, it's important to think about the audience. distinct audiences have distinct needs. Recruiters want to hear about importance and impact right

away, while technical peers care about correctness and completeness. I had to simplify when I had to, consider the pros and cons of how much to explain against how much to let the statistics do the talking, and it took me time to build a solution that could do both. This article completes my AI/ML portfolio by combining model creation, data processing, communication, and now ethical assessment (Wiśniewski & Biecek, 2022). By showing all four artifacts together, it gives better context of my AI and ML skills.

References

- Alelyani, S. (2021). Detection and evaluation of machine learning bias. *Applied Sciences*, 11(14), 6271. <https://www.mdpi.com/2076-3417/11/14/6271>
- Wiśniewski, J., & Biecek, P. (2022). fairmodels: a Flexible Tool for Bias Detection, Visualization, and Mitigation in Binary Classification Models. *R Journal*, 14(1).
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